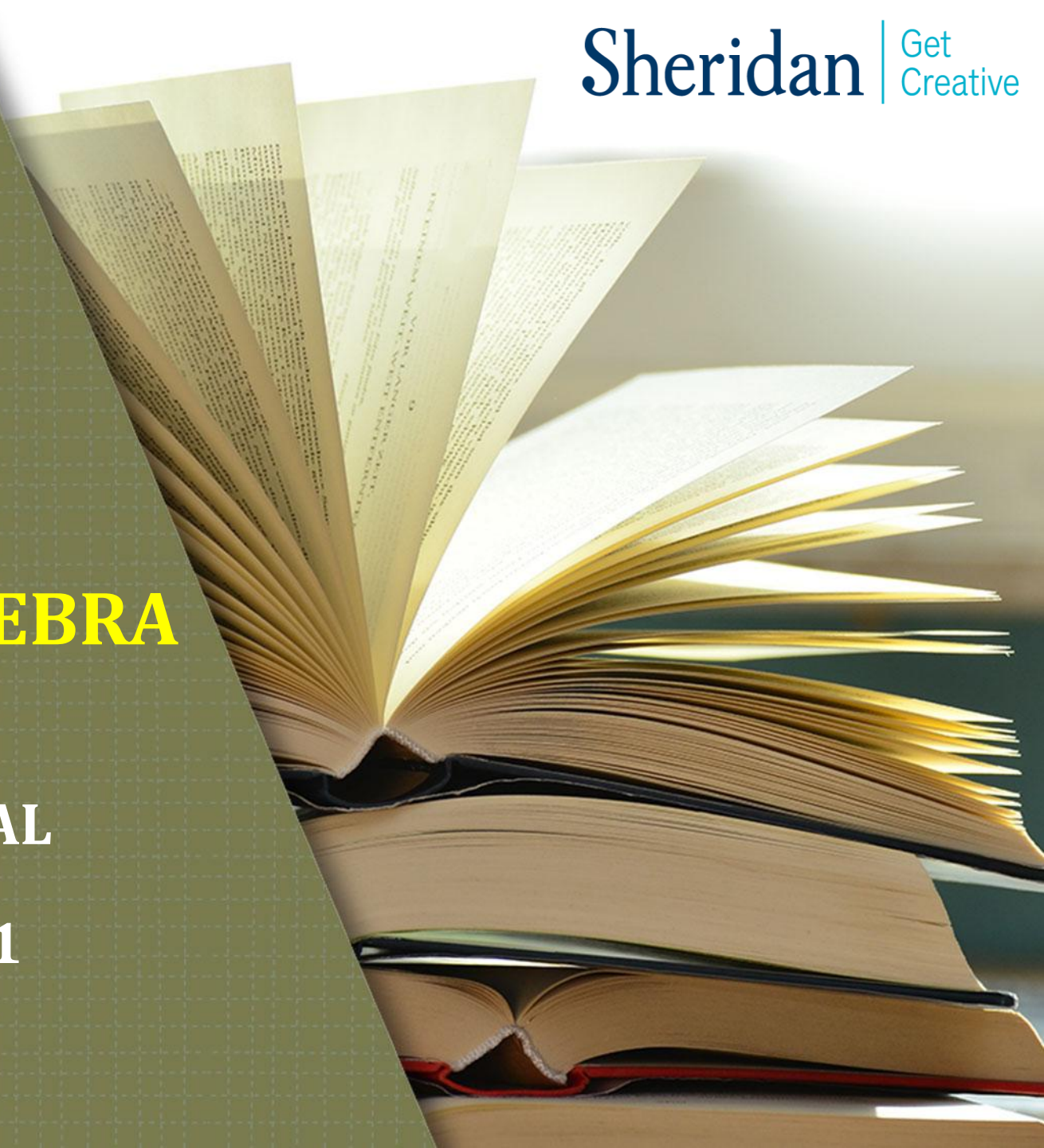


LINEAR ALGEBRA

TUTORIAL

WEEK 1



Complex Numbers

$$z = a + ib \quad i = \sqrt{-1} \quad i^2 = -1$$

a : real part

b : imaginary part

$$\bar{z} = a - ib$$

Complex Conjugate of z

$$|z| = \sqrt{z\bar{z}} = \sqrt{a^2 + b^2}$$

$$z = |z|(\cos\varphi + i\sin\varphi)$$

$$z = |z|e^{i\varphi}$$

$$a = |z|\cos\varphi \quad \text{and} \quad b = |z|\sin\varphi$$

$$\varphi = \tan^{-1} \frac{b}{a}$$

$$-\pi < \varphi < \pi$$

$|z|$ is Modulus of z

φ is Argument of z

Operation with Complex Numbers

$$z_1 = a + ib$$

$$z_2 = c + id$$

Addition : $z_1 + z_2 = (a + ib) + (c + id) = (a + c) + i(b + d)$

Subtraction : $z_1 - z_2 = (a + ib) - (c + id) = (a - c) + i(b - d)$

Multiplication: $z_1 z_2 = (a + ib)(c + id) = ac + i ad + i bc + i^2 bd$
 $= (ac - bd) + i (ad + bc)$

Division : $z_1/z_2 = (a + ib)/(c + id)$

$$= \frac{a + ib}{c + id} \times \frac{c - id}{c - id}$$
$$= \frac{(ac + bd) + i(bc - ad)}{c^2 - i^2 d^2}$$
$$= \frac{(ac + bd) + i(bc - ad)}{c^2 + d^2}$$

$$z_1 z_2 = |z_1| |z_2| [\cos(\varphi_1 + \varphi_2) + i \sin(\varphi_1 + \varphi_2)]$$

$$z_1 z_2 = |z_1 z_2| e^{i(\varphi_1 + \varphi_2)}$$

$$\frac{z_1}{z_2} = \frac{|z_1|}{|z_2|} [\cos(\varphi_1 - \varphi_2) + i \sin(\varphi_1 - \varphi_2)]$$

$$\frac{z_1}{z_2} = \left| \frac{z_1}{z_2} \right| e^{i(\varphi_1 - \varphi_2)}$$

Moivre's Formula:

$$z^n = |z|^n (\cos n\varphi + i \sin n\varphi)$$

$$z^n = |z|^n e^{in\varphi}$$

$$z^{\frac{1}{n}} = |z|^{\frac{1}{n}} \left[\cos \left(\frac{\theta}{n} + \frac{2k\pi}{n} \right) + i \sin \left(\frac{\theta}{n} + \frac{2k\pi}{n} \right) \right] = |z|^{\frac{1}{n}} e^{i \frac{\theta + 2k\pi}{n}}$$

The number a is called the ***real part*** of z and is denoted by $\operatorname{Re}(z)$, and the number b is called the ***imaginary part*** of z and is denoted by $\operatorname{Im}(z)$. Thus,

$$\operatorname{Re}(3 + 2i) = 3, \quad \operatorname{Im}(3 + 2i) = 2$$

$$\operatorname{Re}(1 - 5i) = 1, \quad \operatorname{Im}(1 - 5i) = \operatorname{Im}(1 + (-5)i) = -5$$

$$\operatorname{Re}(7i) = \operatorname{Re}(0 + 7i) = 0, \quad \operatorname{Im}(7i) = \operatorname{Im}(0 + 7i) = 7$$

$$\operatorname{Re}(4) = \operatorname{Re}(4 + 0i) = 4, \quad \operatorname{Im}(4) = \operatorname{Im}(4 + 0i) = 0$$

Find the conjugate and modulus for each of these numbers

$$z = i, \quad z = 1, \quad z = -4 - 5i, \quad z = 3 + 4i$$

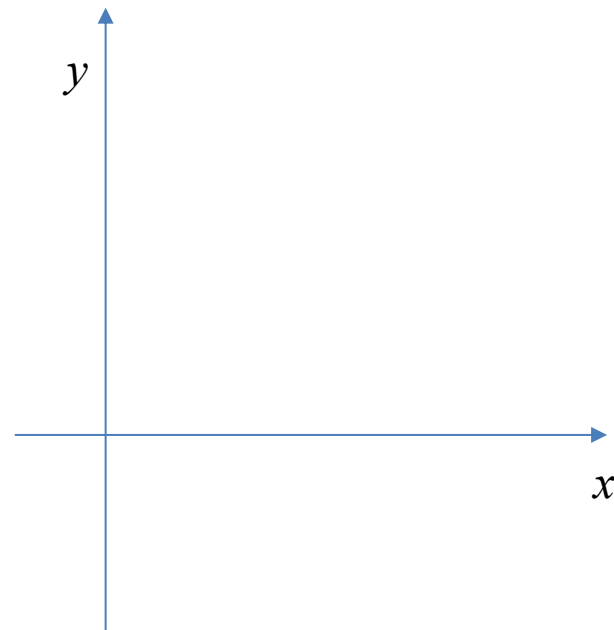
Show the numbers below on the complex plane

$$z_1 = 3 + 4i, \quad z_2 = 3 - 4i, \quad z_3 = -3 + 4i, \quad z_4 = -3 - 4i$$

$$z_5 = 1, \quad z_6 = -1, \quad z_7 = i, \quad z_8 = -i,$$

$$z_9 = 1 + i, \quad z_{10} = 1 - i, \quad z_{11} = -1 + i, \quad z_{12} = -1 - i$$

$$z_{13} = 3 + 4i \quad \text{and} \quad z_{14} = 4 + 3i$$



Find the multiplication of the complex numbers below
 $z_1 z_2$ and division $\frac{z_1}{z_2}$

$$z_1 = 2 - 5i \quad , \quad z_2 = 4 + 3i$$

$$z_1 = \sqrt{5} - 5i \quad , \quad z_2 = 2 + i$$

Using $z_1 = 2 - 5i$, and $z_2 = 4 + 3i$, explore the application of the properties below

THEOREM 1 The following results hold for any complex numbers z , z_1 , and z_2 .

- 1) $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$
- 2) $\overline{z_1 - z_2} = \overline{z_1} - \overline{z_2}$
- 3) $\overline{z_1 z_2} = \overline{z_1} \overline{z_2}$
- 4) $\overline{z_1 / z_2} = \overline{z_1} / \overline{z_2}$
- 5) $\overline{\overline{z}} = z$

THEOREM 2 The following results hold for any complex numbers z , z_1 , and z_2 .

- 1) $|\overline{z}| = |z|$
- 2) $|z_1 z_2| = |z_1| |z_2|$
- 3) $|z_1 / z_2| = |z_1| / |z_2|$
- 4) $|z_1 + z_2| \leq |z_1| + |z_2|$

Express $z = 1 - \sqrt{3}i$ in polar form using the principal argument.

Solution The modulus of z is

$$|z| = \sqrt{1^2 + (-\sqrt{3})^2} = \sqrt{4} = 2$$

Thus, it follows from (10) with $a = 1$ and $b = -\sqrt{3}$ that

$$1 = 2 \cos \phi \quad \text{and} \quad -\sqrt{3} = 2 \sin \phi$$

and this implies that

$$\cos \phi = \frac{1}{2} \quad \text{and} \quad \sin \phi = -\frac{\sqrt{3}}{2}$$

The unique angle ϕ that satisfies these equations and whose radian measure satisfies (12) is $\phi = -\pi/3$ (**Figure B.7**). Thus, a polar form of z is

$$z = 2\left(\cos\left(-\frac{\pi}{3}\right) + i \sin\left(-\frac{\pi}{3}\right)\right) = 2\left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3}\right)$$

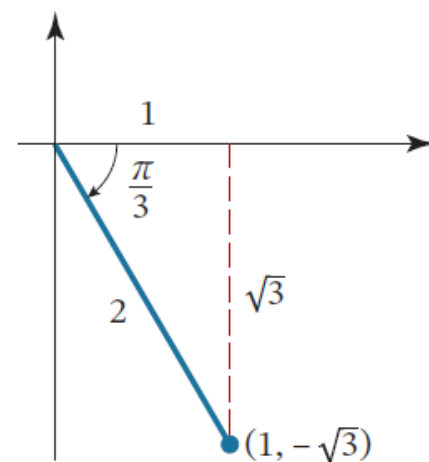


FIGURE B.7

Do the same conversion to $z_1 = 1 + \sqrt{3}i$ and $z_2 = \sqrt{3} + i$

Using the numbers: $z_1 = 2 + 2i$, $z_2 = 1 - \sqrt{3}i$

Find the following amounts in polar forms

$$z_1 + z_2$$

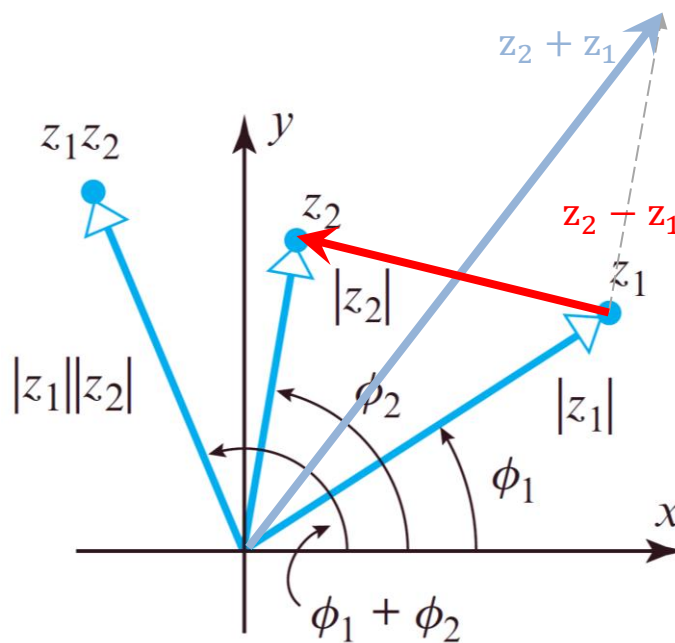
$$z_2 - z_1$$

$$z_1 z_2$$

$$\frac{z_1}{z_2}$$

$$z_2$$

$$iz_1$$



Using De Moivre's Formula, find the following:

$$z_1 = (1 + i)^3, \quad z_2 = (1 - \sqrt{3}i)^2$$

$$z_3 = (1 + i)^{1/3}, \quad z_4 = \sqrt{1 - \sqrt{3}i}$$

$$z_5 = \sqrt{(1 - \sqrt{3}i)^5}, \quad z_6 = \sqrt[3]{-1}$$

Find the following amounts using Euler formula:

$$1) e^{i\pi/2} + e^{i\pi}$$

$$2) e^{2+i\pi}$$

$$3) e^{2+i\pi} e^{3+\frac{i\pi}{3}}$$

$$4) e^{2+i\pi} / e^{3+\frac{i\pi}{3}}$$

1) $z = -3 + 2j$ $z^4 = ?$

2) $z = 3 - 4j$ $z^6 = ?$

3) $z = -2 - 6j$ $z^4 = ?$

4) $z=5+4j$ $z^3=?5$

5) $z=4e^{j\pi/2}$ $z^2=?$

6- Find the third roots
of $2\sqrt{11} + 10j$

$$z^{\frac{1}{n}} = |z|^{\frac{1}{n}} \left[\cos \left(\frac{\theta}{n} + \frac{2k\pi}{n} \right) + i \sin \left(\frac{\theta}{n} + \frac{2k\pi}{n} \right) \right]$$

$$Z = 2\sqrt{11} + 10j$$

$$Z^{\frac{1}{3}} = ?$$

$$Z = 2\sqrt{11} + 10j = r (\cos \theta + j \sin \theta)$$

$$r = \sqrt{4 \times 11 + 100} = \sqrt{144} = 12$$

$$\theta = \tan^{-1} \frac{10}{2\sqrt{11}} \approx 56.4^\circ$$

$$W_0 = \sqrt[3]{12} \left[\cos \left(\frac{56.4}{3} + \frac{0 \times 360}{3} \right) + j \sin \left(\frac{56.4}{3} + \frac{0}{3} \right) \right]$$

$$\approx 2.1673 + 0.7378j$$

$$W_1 = \sqrt[3]{12} \left[\cos \left(\frac{56.4}{3} + \frac{360}{3} \right) + j \sin \left(\frac{56.4}{3} + \frac{360}{3} \right) \right]$$

$$\approx -1.7230 + 1.5076j$$

$$W_2 = \sqrt[3]{12} \left[\cos \left(\frac{56.4}{3} + \frac{720}{3} \right) + j \sin \left(\frac{56.4}{3} + \frac{720}{3} \right) \right]$$

$$\approx -0.4447 - 2.245j$$



7-Solve the equation of $(z + i)^n + (z - i)^n = 0$.



Solution -Solve the equation of $(z + i)^n + (z - i)^n = 0$.

$$\left(\frac{z+i}{z-i}\right)^n = -1 \quad \text{Let } w = \frac{z+i}{z-i}. \text{ The equation becomes } w^n = -1 = e^{i\pi}$$

Using Euler's formula, the solutions for w are $w_k = e^{i\frac{(2k+1)\pi}{n}}$ for $k = 0, 1, 2, \dots, n-1$

$$w(z-i) = z+i$$

$$wz - iw = z + i$$

$$z(w-1) = i(w+1)$$

$$z = i\left(\frac{w+1}{w-1}\right)$$

$$\text{Substitute } w_k = e^{i\theta}, \quad z = i\left(\frac{e^{i\theta} + 1}{e^{i\theta} - 1}\right)$$

$$\begin{aligned} \text{Factor out } e^{i\theta/2} \quad z &= i\left(\frac{e^{i\theta/2}(e^{i\theta/2} + e^{-i\theta/2})}{e^{i\theta/2}(e^{i\theta/2} - e^{-i\theta/2})}\right) \\ &= i\left(\frac{2\cos(\theta/2)}{2i\sin(\theta/2)}\right) = \cot\left(\frac{\theta}{2}\right) \end{aligned}$$

$$\text{Therefore } z_k = \cot\left(\frac{(2k+1)\pi}{2n}\right) \quad \text{for } k = 0, 1, 2, \dots, n-1$$

8-If $I = e^{ix_1} e^{ix_2} e^{ix_3} \dots e^{ix_n}$, find the following items.

1) $\sin(x_1 + x_2 + \dots + x_n)$

2) $\cos(x_1 + x_2 + \dots + x_n)$

3) $\tan(x_1 + x_2 + \dots + x_n)$

